



Exercise 5

Goal of the exercise : 1) Performance analysis of a system with intersymbol interference. 2) Different notions of entropy.

Exercise 1

Let $\{s_n\}_{n \geq 0}$ be a sequence of i.i.d. symbols taking value in $\{-A, A\}$ equiprobably. This sequence is sent over a channel so that at each discrete-time $n \geq 0$, the receiver observes

$$y_n = s_n + \alpha s_{n-1} + w_n, \quad (1)$$

where

- $\{w_n\}$ is an i.i.d. sequence of real Gaussian noises with zero mean and variance $N_0/2$,
- α is a real constant between 0 and 1, and
- $s_{-1} = 0$.

Typically, the term αs_{n-1} in equation (1) corresponds to an echo and α to its attenuation.

We study the successive interference cancellation decoder, which decodes the symbols according to the following rules :

$$\hat{s}_0 = A \cdot \text{sign}(y_0) \quad (2)$$

and

$$\hat{s}_\ell = A \cdot \text{sign}(z_\ell), \quad \ell = 1, 2, \dots, \quad (3)$$

where

$$z_\ell := y_\ell - \alpha \cdot \hat{s}_{\ell-1}.$$

The receiver thus consecutively decodes the symbols $s_0, s_1, \dots$ and when it decodes s_ℓ , it uses its decision $\hat{s}_{\ell-1}$ to eliminate the interference caused by $s_{\ell-1}$. Notice that if $\hat{s}_{\ell-1} = s_{\ell-1}$, then $z_\ell = s_\ell + w_\ell$ and thus the “interference” has disappeared.

Questions :

1. Express $P_{e,0} := \Pr[\hat{s}_0 \neq s_0]$ in terms of E_b/N_0 .
2. Consider now an arbitrary $\ell = 1, 2, \dots$ and assume that the previous decision of symbol $s_{\ell-1}$ was correct, i.e. :

$$\hat{s}_{\ell-1} = s_{\ell-1}. \quad (4)$$

Express $\Pr[\hat{s}_\ell \neq s_\ell | \hat{s}_{\ell-1} = s_{\ell-1}]$ in fonction of E_b/N_0 and α .

3. Assume now that

$$\hat{s}_{\ell-1} \neq s_{\ell-1}. \quad (5)$$

Express z_ℓ in terms of α, s_n, s_{n-1} and w_n when (5) holds.

4. Express $\Pr[\hat{s}_\ell \neq s_\ell | \hat{s}_{\ell-1} \neq s_{\ell-1}]$ in fonction of E_b/N_0 and α .

Hint : Find $\Pr[\hat{s}_\ell \neq s_\ell | \hat{s}_{\ell-1} \neq s_{\ell-1}, s_{\ell-1} = A]$ and $\Pr[\hat{s}_\ell \neq s_\ell | \hat{s}_{\ell-1} \neq s_{\ell-1}, s_{\ell-1} = -A]$

5. Use the answers from questions 2. and 4. to express $P_{e,\ell} := \text{Prob}(\hat{s}_\ell \neq s_\ell)$ as a function of $P_{e,\ell-1}$, E_b/N_0 , and α , for all $\ell = 1, 2, \dots$

6. Express $P_{e,\ell}$ as a function of α , ℓ , and E_b/N_0 .

Hint : Find the solution to the sequence defined in the previous question.

7. Keeping only the dominant term in $P_{e,\ell}$ at high SNR, approximate the error probability using a single term.

Hint : $Q(a) \ll 1$ for $a \gg 1$.

8. We want to compare the error probability approximation obtained in the previous question to that of a threshold detector that does not take interference into account and directly decides as follows :

$$\tilde{s}_\ell = A \cdot \text{sign}(y_\ell), \quad \ell = 0, 1, 2, \dots \quad (6)$$

It can be shown that the error probability of the decision rule in (6) is given by $\tilde{P}_e \approx \frac{1}{2} Q\left(\sqrt{2(1-\alpha)^2 \frac{E_b}{N_0}}\right)$.

What is the SNR gain obtained by using the decision-feedback decoder in (3) ?

Exercise 2

Let the random variable X be uniformly distributed over the set $\{1, 2, 3, 4, 5, 6\}$ and

$$\begin{aligned} Y &= X \bmod 2 \\ Z &= X \bmod 3. \end{aligned}$$

1. Draw a table that indicates the value of the pair (Y, Z) for each value of $X \in \{1, \dots, 6\}$.
2. Find the entropies $H(X)$, $H(Y)$ et $H(Z)$.
3. Find the conditional entropies $H(X|Y, Z)$ et $H(X|Y)$.
4. Find the joint entropies $H(Y, Z)$ et $H(X, Y)$.
5. Find the conditional entropy $H(Y|Z)$. Compare it to $H(Y)$. What do you deduce ?